

Exercise Sheet #2: Basic Concepts 2

Lecturer: Prof. Wick
Due: 20 Oct 2025

Exercise 1: Resistance of Wires

Given are two copper wires of equal mass with circular cross-sections; wire A is twice as long as wire B. How do the resistances of the wires compare (neglecting temperature effects)?

- a) $R_A = 8R_B$,
- b) $R_A = 4R_B$
- c) $R_A = 2R_B$,
- d) $R_A = R_B$.

Solution

The resistance of a wire is given by

$$R = \rho \frac{l}{A},$$

where ρ is the resistivity, l is the length, and A is the cross-sectional area. Since both wires have the same mass and material, we have

$$m = \rho_m V = \rho_m (A \cdot l) = \text{const.}$$

Thus, $A \cdot l = \text{const.}$, which implies $A \propto \frac{1}{l}$.

For wire A, the length is $l_A = 2l_B$. Therefore, its cross-sectional area is

$$A_A = \frac{1}{2} A_B.$$

Now the resistance ratio becomes

$$\frac{R_A}{R_B} = \frac{\rho \frac{l_A}{A_A}}{\rho \frac{l_B}{A_B}} = \frac{l_A}{l_B} \cdot \frac{A_B}{A_A} = \frac{2l_B}{l_B} \cdot \frac{A_B}{\frac{1}{2}A_B} = 2 \cdot 2 = 4.$$

Therefore,

$$R_A = 4R_B.$$

Exercise 2: Resistivity

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Solution

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Exercise 3: Temperature Dependence of Resistance

At what temperature does a copper wire have a resistance that is 10% greater than at 20 °C?

Solution

The temperature dependence of the resistance is given by

$$R(T) = R_0(1 + \alpha(T - T_0)),$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

We are looking for the temperature T at which

$$R(T) = 1.1 R_0.$$

Thus,

$$1.1R_0 = R_0(1 + \alpha(T - 20^\circ\text{C})).$$

Dividing through by R_0 :

$$1.1 = 1 + \alpha(T - 20).$$

Hence,

$$T - 20 = \frac{0.1}{\alpha}.$$

For copper, $\alpha \approx 0.0039 \text{ K}^{-1}$, so

$$T \approx 20 + \frac{0.1}{0.0039} \approx 45.6^\circ\text{C}.$$

Therefore, the copper wire has a 10% greater resistance at approximately

$$\boxed{46^\circ\text{C}}$$

than at 20 °C.

Exercise 4: Energy and Power

A particle accelerator produces a proton beam with a current of $3.50 \mu\text{A}$. Each proton has an energy of 60.0 MeV. The protons strike a target in a vacuum chamber and come to rest in it, heating the target.

- How many protons strike the target per second?
- How much energy is deposited in the target per second?

Solution

The beam current is $I = 3.50 \mu\text{A} = 3.50 \times 10^{-6} \text{ A}$. The elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$.

- a) The number of protons per second is

$$\dot{N} = \frac{I}{e}.$$

Substituting,

$$\dot{N} = \frac{3.50 \times 10^{-6}}{1.602176634 \times 10^{-19}} \approx 2.18 \times 10^{13} \text{ s}^{-1}.$$

- b) Each proton has an energy of 60.0 MeV.

The deposited energy per second is

$$P = U \cdot I = \frac{W}{Q} \cdot I = \frac{60.0 \text{ MeV}}{e} \cdot 3.50 \mu\text{A} = 210 \text{ W}.$$